

Ratio: answers and worked method

Answers to step-by-step practice 1

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	An 84 ml paint mix contains red and white paint in the ratio 3:4. How much red paint is in the mix?	Notebooks and folders are in the ratio 4:7. There are 35 folders. How many notebooks are there?	Adults and children are in the ratio 2:5. There are 24 more children than adults. How many people are there altogether?
Keywords & Calculation	Total known: $3 + 4 = 7$ parts. Find 1 part, then red paint's 3 parts.	35 folders is 7 parts. Find 1 part, then 4 parts for notebooks.	24 more children is $5 - 2 = 3$ parts. Find 1 part, then 7 parts altogether.
Written Method	$84 \text{ ml} \div 7 = 12 \text{ ml per part. } 12 \text{ ml} \times 3 = 36 \text{ ml.}$ Red White 84 ml altogether 1 part = 12 ml	$35 \div 7 = 5 \text{ per part. } 5 \times 4 = 20 \text{ notebooks.}$ Notebooks Folders Folders: 35 1 part = 5 items	$24 \div 3 = 8 \text{ per part. } 8 \times 7 = 56 \text{ people.}$ Adults Children 3 extra parts = 24 people 1 part = 8 people
Ballpark Check & Math	36 ml is less than half of 84 ml. The other share is 48 ml; together they make 84 ml.	4 parts is just over half of 7 parts, so expect just over half of 35. 20 fits.	The total is 7 parts, the difference 3. Expect between 2×24 and 3×24 : 48 to 72. 56 fits.
Final Answer	Red paint: 36 ml.	20 notebooks.	56 people altogether.

Ratio: check your method

Step-by-step practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 7 parts; $84\text{ ml} \div 7 = 12\text{ ml per part}$; $12\text{ ml} \times 3 = 36\text{ ml of red paint}$.	Method 35 folders represent 7 parts. $35 \div 7 = 5\text{ per part}$; $5 \times 4 = 20\text{ notebooks}$.	Method The difference is $5 - 2 = 3\text{ parts}$. $24 \div 3 = 8\text{ per part}$; $8 \times 7 = 56\text{ people}$.
If you got... 48 ml: check whether you used $12\text{ ml} \times 4$. That is white paint; red has 3 parts.	If you got... 32 notebooks: check whether you used $35 - (7 - 4)$. The difference of 3 is in parts, not individual items.	If you got... 168 people: check whether you used 24×7 . The 24 represents 3 parts, not 1; divide by 3 first.
Check Red: 36 ml; White: 48 ml. $36\text{ ml} + 48\text{ ml} = 84\text{ ml}$, and 36:48 simplifies to 3:4.	Check 20:35 simplifies to 4:7. There should be fewer notebooks than folders.	Check Adults: 16; children: 40. $40 - 16 = 24$; $16 + 40 = 56$.

Ratio: answers and worked method

Answers to step-by-step practice 2

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A 96 cm ribbon is cut into pieces A and B in the ratio 5:7. Find the length of B.	Juice and water are mixed in the ratio 2:7. There are 18 litres of juice. How many litres of water are needed?	Two lengths of ribbon, A and B, are in the ratio 5:8. B is 21 cm longer than A. What is their total length?
Keywords & Calculation	Total known: $5 + 7 = 12$ parts. Find 1 part, then B's 7 parts.	18 litres of juice is 2 parts. Find 1 part, then 7 parts of water.	21 cm is $8 - 5 = 3$ parts. Find 1 part, then 13 parts altogether.
Written Method	96 cm $\div$ 12 = 8 cm per part. 8 cm $\times$ 7 = 56 cm. A B 96 cm altogether 1 part = 8 cm	18 $\div$ 2 = 9 litres per part. 9 $\times$ 7 = 63 litres. Juice Water Juice: 18 litres 1 part = 9 litres	21 $\div$ 3 = 7 cm per part. 7 $\times$ 13 = 91 cm. A B 3 extra parts = 21 cm 1 part = 7 cm
Ballpark Check & Math	B is more than half of 96 cm. 56 cm fits; 40 cm + 56 cm = 96 cm.	Water is between 3 and 4 times the juice: between 54 and 72 litres. 63 fits.	The total is just over 4 times the 21 cm difference. 4 $\times$ 21 = 84; 91 is reasonable.
Final Answer	B is 56 cm.	63 litres of water.	91 cm altogether.

Ratio: check your method

Step-by-step practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 12 parts; $96\text{ cm} \div 12 = 8\text{ cm}$ per part; $8\text{ cm} \times 7 = 56\text{ cm}$ for B.	Method 18 litres represent 2 parts. $18 \div 2 = 9$ per part; $9 \times 7 = 63$ litres of water.	Method The difference is $8 - 5 = 3$ parts. $21 \div 3 = 7\text{ cm}$ per part; $7 \times 13 = 91\text{ cm}$.
If you got... 40 cm: check whether you used $8\text{ cm} \times 5$. That is piece A; the question asks for B.	If you got... 23 litres: check whether you used $18 + (7 - 2)$. Five extra parts do not mean five extra litres.	If you got... 273 cm: check whether you used 21×13 . The 21 cm represents 3 parts, not 1.
Check $40\text{ cm} + 56\text{ cm} = 96\text{ cm}$. B is more than half of 96 cm.	Check 18:63 simplifies to 2:7. Water is between 3 and 4 times the juice: 54 to 72 litres.	Check Lengths: 35 cm and 56 cm. Difference: 21 cm. Total: 91 cm.

Ratio: answers

Independent practice 1 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £36	1. 300 g	1. 24 m
2. 120 ml	2. 24 tickets	2. Red: 6 counters Blue: 30 counters
3. Shorter: 40 cm Longer: 56 cm	3. 20 m	3. £40
4. 30 beads	4. £63	4. 675 g
5. 250 g	5. 18 L	5. Water: 1200 ml Juice: 200 ml
6. 18 min	6. 40 min	6. 99 min

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $2 + 3 = 5$ parts. $£60 \div 5 = £12$ per part. Jo gets $£12 \times 3 = £36$.	Method Q3: $7 - 4 = 3$ parts. $15 \div 3 = 5$ m per part. Rope A: $5 \times 4 = 20$ m.	Method Q3: $7 + 3 = 10$ parts. $£100 \div 10 = £10$ per part. Shares £70 and £30; difference £40.
If you got... Q1 — £24: check whether you used $£12 \times 2$. That is Sam's share; Jo has 3 parts.	If you got... Q3 — 60 m: check whether you used 15×4 . The 15 m difference represents 3 parts, not 1.	If you got... Q3 — £70: check whether you stopped at $£10 \times 7$. That is Ava's share; subtract Leo's £30.
Check Q1: $£24 + £36 = £60$; $24:36 = 2:3$.	Check Q3: A is 20 m; B is 35 m. Difference 15 m; ratio 4:7.	Check Q3: $£70 + £30 = £100$; $£70 - £30 = £40$.

Ratio: answers

Independent practice 2 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 70 g	1. 27 km	1. 440 g
2. A: 63 m B: 54 m	2. 210 ml	2. 25 L
3. 30 min	3. £35	3. Reading: 24 min Writing: 54 min
4. £40	4. 72 min	4. 63 cm
5. 35 flowers	5. 140 g	5. £26
6. 240 ml	6. 80 animals	6. 153 tickets

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q4: Kim’s £56 represents 7 parts. $£56 \div 7 = £8$ per part. Ali gets $£8 \times 5 = £40$.	Method Q2: $3 + 4 = 7$ parts. $490 \div 7 = 70$ ml per part. Red paint: $70 \times 3 = 210$ ml.	Method Q2: Tank A’s 15 litres is 3 parts. $15 \div 3 = 5$ per part. Tank B: $5 \times 8 = 40$ litres. Difference: $40 - 15 = 25$ litres.
If you got... Q4 — £54: check whether you used $56 - (7 - 5)$. Two fewer parts mean £16 less, not £2 less.	If you got... Q2 — 70 ml: check whether you stopped after $490 \div 7$. That is one part; red paint needs 3 parts.	If you got... Q2 — 40 litres: check whether you stopped at 5×8 . The question asks how much more, so subtract A’s 15 litres.
Check Q4: $£56:£40 = 7:5$. Both shares give £8 per part.	Check Q2: Red 210 ml + white 280 ml = 490 ml; ratio 3:4.	Check Q2: $15:40 = 3:8$, and $15 + 25 = 40$ litres.

Ratio: answers

Independent practice 3 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 24 members	1. 72 min	1. 8 min
2. 400 ml	2. 56 mm	2. £168
3. Dan: £14 Pat: £84	3. 210 g	3. 77 animals
4. 90 min	4. 24 beads	4. 150 ml
5. 100 g	5. £9	5. 91 m
6. 72 cm	6. 825 ml	6. Oats: 500 g Raisins: 450 g

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 3 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $7 + 3 = 10$ parts. $80 \div 10 = 8$ per part. Seniors: $8 \times 3 = 24$ members.	Method Q3: Butter's 90 g is 3 parts. $90 \div 3 = 30$ g per part. Total: $30 \times (4 + 3) = 210$ g.	Method Q2: May's £60 is 5 parts. $\pounds 60 \div 5 = \pounds 12$ per part. Total: $\pounds 12 \times (9 + 5) = \pounds 168$.
If you got... Q1 — 56: check whether you used 8×7 . That counts junior members, not seniors.	If you got... Q3 — 120 g: check whether you stopped at 30×4 . That is flour only; include 90 g of butter.	If you got... Q2 — £108: check whether you stopped at $\pounds 12 \times 9$. That is Kai's share; add May's £60.
Check Q1: 56 juniors + 24 seniors = 80 members, with ratio 7:3.	Check Q3: Flour 120 g + butter 90 g = 210 g; ratio 4:3.	Check Q2: $\pounds 108 + \pounds 60 = \pounds 168$; $108:60 = 9:5$.

Ratio: answers

Independent practice 4 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. 20 min	1. £192	1. 9 L
2. First: 5 L Second: 15 L	2. 54 plants	2. 540 g
3. 36 cm	3. 5 min	3. 45 cm
4. £84	4. 12 kg	4. 117 tickets
5. Dark: 12 chocolates Milk: 48 chocolates	5. 21 L	5. £88
6. 240 g	6. 21 m	6. 50 min

Check what is requested: one amount, both amounts, a total or a difference.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 4 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q4: Will's £48 is 4 parts. $£48 \div 4 = £12$ per part. Noor gets $£12 \times 7 = £84$.	Method Q4: $7 - 3 = 4$ parts. $16 \div 4 = 4$ kg per part. Lighter sack: $4 \times 3 = 12$ kg.	Method Q6: $7 + 2 = 9$ parts. $90 \div 9 = 10$ minutes per part. Work 70; breaks 20. Difference: 50 minutes.
If you got... Q4 — £12: check whether you stopped after $£48 \div 4$. Noor receives 7 parts, not 1.	If you got... Q4 — 48 kg: check whether you used 16×3 . The 16 kg difference represents 4 parts, not 1.	If you got... Q6 — 70 minutes: check whether you stopped at 10×7 . Subtract 20 minutes of breaks to find how much longer.
Check Q4: $£84:£48 = 7:4$. Both shares give £12 per part.	Check Q4: Heavier sack 28 kg; lighter 12 kg. Difference 16 kg; ratio 7:3.	Check Q6: $70 + 20 = 90$ minutes; $70 - 20 = 50$ minutes.